Tharunkumar Chandrasekaran

+1 (240) 350-9147 | tharun2k1@gmail.com | LinkedIn | GitHub | Hyattsville, MD

AI Software Engineer experienced in building AI-powered applications, enterprise workflows, and scalable backend services using Python, Django, AWS, and LLM APIs, with expertise in workflow automation and a Master's in Data Science.

Experience

AI Software Engineer June 2026 – Present
Jansy Packaging *Englewood Cliffs, NJ*
- Developed enterprise ERP modules for sales budgeting and manufacturing inspections using Django, Vue.js, and REST APIs, supporting operational workflows.
- Built backend APIs and data workflows for sales budgeting, inspections, and operational reporting in Slate ERP.
- Developed an AI Inspection Assistant that analyzes inspection data and generates automated inspection reports, integrating AI capabilities into manufacturing inspection workflows.

Graduate Research Assistant September 2025 – May 2026
Digital Engagement Lab, University of Maryland *College Park, MD*
- Conducted data-driven research analyzing digital engagement and attention patterns across short-form media.
- Developed an AI-powered News Summarization Platform using Next.js and TypeScript, integrating the Anthropic Claude API for automated content extraction, summarization, and analysis.

Data Analytics Intern June 2025 – August 2025
Greater New York Insurance *New York City, NY*
- Built scalable Python and SQL pipelines to ingest, transform, and analyze large-scale NYC 311 datasets for risk hotspot analysis.
- Designed optimized SQL workflows for data extraction, validation, and feature engineering, improving data quality and model readiness.
- Developed a Snowflake data platform and Django application to deliver risk insights through interactive dashboards.

Software Engineer March 2022 – August 2024
Freshworks *Chennai, India*
- Migrated enterprise onboarding workflows and business rules to a multi-workspace architecture, enabling scalable workflow automation for customers.
- Revamped Freshservice user profiles and updated REST API endpoints to maintain compatibility during the platform's API v1 deprecation.
- Built GitHub Actions CI/CD pipelines for automated testing and containerized deployments; developed employee off-boarding workflows and improved regression test pass rates from 72% to 96%.

Projects

Bitcoin Price Prediction Pipeline | *Apache Spark, Docker, AWS, NeuralProphet*
- Built a real-time Apache Spark pipeline using Docker to ingest Bitcoin prices from the CoinGecko API and compute moving-average metrics.
- Deployed data processing and storage on AWS (EC2, S3) with InfluxDB, and forecasted price trends using NeuralProphet.

Smart Story Suite – AI News Summarization Platform | *Next.js, TypeScript, Claude API*
- Built a Next.js and TypeScript platform that extracts news content through REST APIs and converts articles into structured insights.
- Integrated the Anthropic Claude API to generate summaries, highlights, and article scores with real-time visualization.

NYC 311 Risk Hotspot Analysis Pipeline | *Python, SQL, Django*
- Built an end-to-end pipeline to ingest, clean and analyze large-scale NYC 311 data for risk hotspot analysis.
- Developed optimized SQL workflows and a Snowflake/Django platform for data validation, feature engineering, and interactive risk dashboards.

Technical Skills

Languages: Python, Java, SQL, TypeScript, R
AI & Automation: LLM API Integration, Prompt engineering, LangChain, Anthropic Claude API, TensorFlow
Backend & Integration: Django, REST APIs, ERP Applications, Data Pipelines, ETL/ELT
Data & Orchestration: Apache Spark, Apache Airflow, HDFS, Snowflake, MySQL, PostgreSQL, MongoDB
Cloud & DevOps: AWS (S3, Glue, Lambda), Azure, Docker, Kubernetes, CI/CD (GitHub Actions), Linux, Git

Education

University of Maryland, College Park August 2024 – May 2026
Master of Science in Data Science, GPA: 3.8/4 *College Park, USA*
Rajalakshmi Engineering College September 2018 – June 2022
Bachelor's in Computer Science and Engineering, GPA: 3.7/4 *Chennai, India*